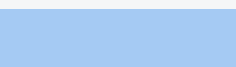
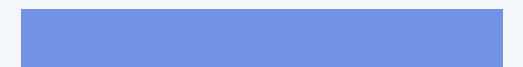
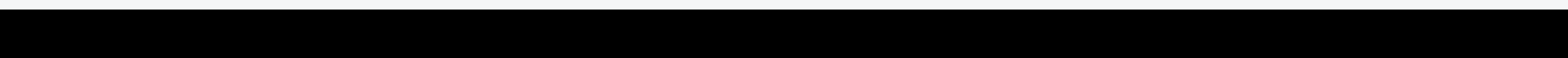
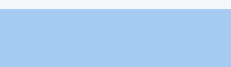
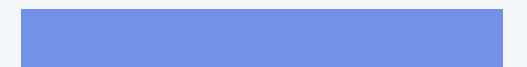
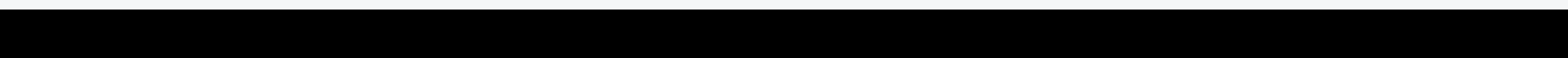


GITHUB AND C++ LIBRARY



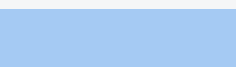
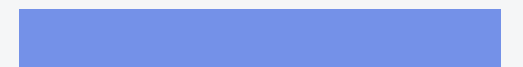
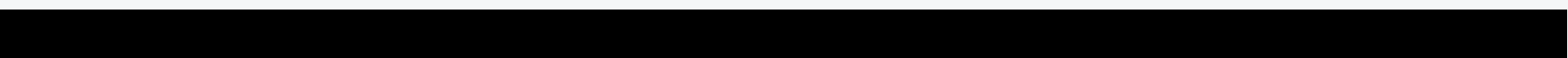
01. Creating a repo
02. Create and publish a library
03. Branching
04. Working on others repo

Agenda



01.

CREATING A REPO

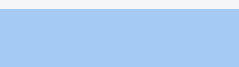
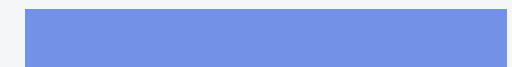
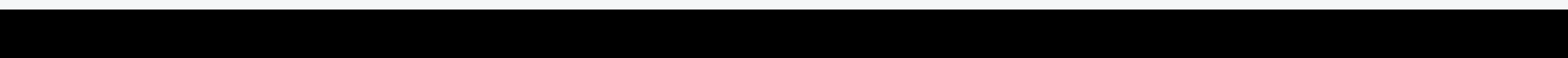


Repository

- A repository is the most basic element of GitHub.
- It's a place where you can store your code, your files, and each file's revision history.
- Repositories can have multiple collaborators and can be either public or private.

Local Repo

Remote Repo



Repository

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk ().*

1

General

Owner *

 shabansatti

▼

Repository name *

/

Great repository names are short and memorable. How about [verbose-waffle](#)?

Description

0 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public

▼

Start with a template

Templates pre-configure your repository with files.

No template

▼

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

☐

○

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

▼

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

▼

Create repository

Repository

Create a new repository

Name

repository name

Description

Local path

D:\UET-LHR\Spring-26\OOP-BS\Test Codes

Choose...

☐ Initialize this repository with a README

Git ignore

None

License

None

Create repository

Cancel

Repository

No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions of what to do next.

Push commits to the origin remote

You have 1 local commit waiting to be pushed to GitHub.

Create a new repository

Name

repository name

Description

Local path

D:\UET-LHR\Spring-26\OOP-BS\Test Codes

Choose...

☐ Initialize this repository with a README

Git ignore

None

License

None

Create repository

Cancel

Repository

Create a new repository

Name

first_library

Description

This is first library repo.

Local path

D:\UET-LHR\Spring-26\OOP-BS

Choose...

☒ Initialize this repository with a README

Git ignore

C++

License

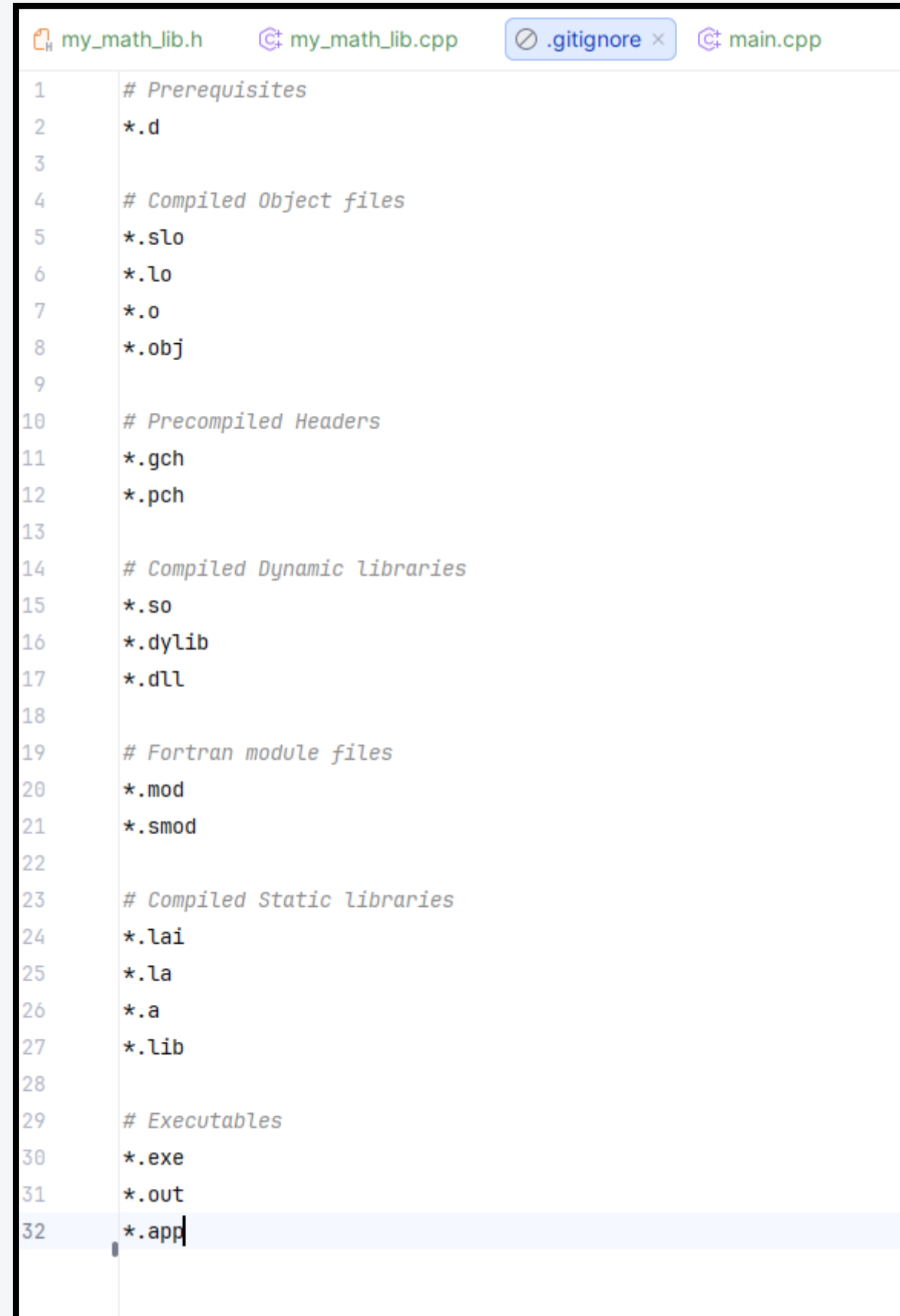
MIT License

The repository will be created at D:\UET-LHR\Spring-26\OOP-BS\first_library.

Create repository

Cancel

gitignore



```
my_math_lib.h  my_math_lib.cpp  .gitignore x  main.cpp
1  # Prerequisites
2  *.d
3
4  # Compiled Object files
5  *.slo
6  *.lo
7  *.o
8  *.obj
9
10 # Precompiled Headers
11 *.gch
12 *.pch
13
14 # Compiled Dynamic Libraries
15 *.so
16 *.dylib
17 *.dll
18
19 # Fortran module files
20 *.mod
21 *.smod
22
23 # Compiled Static libraries
24 *.lai
25 *.la
26 *.a
27 *.lib
28
29 # Executables
30 *.exe
31 *.out
32 *.app
```

Repository

Current repository
first_library

Current branch
main

Publish repository

Publish this repository to GitHub

Changes

History

Filter

☒ 0 changed files

Summary (required)

Description

New

Commit to main

Committed just now
Initial commit

Undo

No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.

Publish your repository to GitHub

This repository is currently only available on your local machine. By publishing it on GitHub you can share it, and collaborate with others.

Always available in the toolbar for local repositories or **Ctrl + P**

Publish repository

Open the repository in your external editor

Select your editor in [Options](#)

Repository menu or **Ctrl + Shift + A**

Open in Visual Studio Code

View the files of your repository in Explorer

Repository menu or **Ctrl + Shift + F**

Show in Explorer

Publish Repo

Publish repository

X

GitHub.com

GitHub Enterprise

Name

first_library

Description

This is first library repo.

☐ Keep this code private

Organization

None

Publish repository

Cancel

shabansatti / first_library

🔍 Type / to search

<> Code

⌚ Issues

🔗 Pull requests

🔄 Actions

📁 Projects

📖 Wiki

🛡 Security

📈 Insights

⚙ Settings

👤 first_library

Public

🌟 Pin

👁 Watch 0

🍴 Fork 0

★ Star 0

🔗 main

🔗 1 Branch

🏷 0 Tags

🔍 Go to file

t

Add file

<> Code

👤 shabansatti

Initial commit

d26c54e · 54 minutes ago

🕒 1 Commit

📄 .gitattributes	Initial commit	54 minutes ago
📄 .gitignore	Initial commit	54 minutes ago
📄 LICENSE	Initial commit	54 minutes ago
📄 README.md	Initial commit	54 minutes ago

📖 README

📄 MIT license

first_library

This is first library repo.

About

⚙

This is first library repo.

📖 Readme

📄 MIT license

📈 Activity

★ 0 stars

👁 0 watching

🍴 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

👤 shabansatti

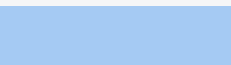
Shaban Satti

Repository

This PC > New Volume (D:) > UET-LHR > Spring-26 > OOP-BS > first_library				
Sort View ...				
Name	Date modified	Type	Size	
 .gitattributes	03/03/2026 10:02 pm	Git Attributes Sour...	1 KB	
 .gitignore	03/03/2026 10:02 pm	Git Ignore Source ...	1 KB	
 LICENSE	03/03/2026 10:02 pm	File	2 KB	
 README.md	03/03/2026 10:02 pm	MD File	1 KB	

02.

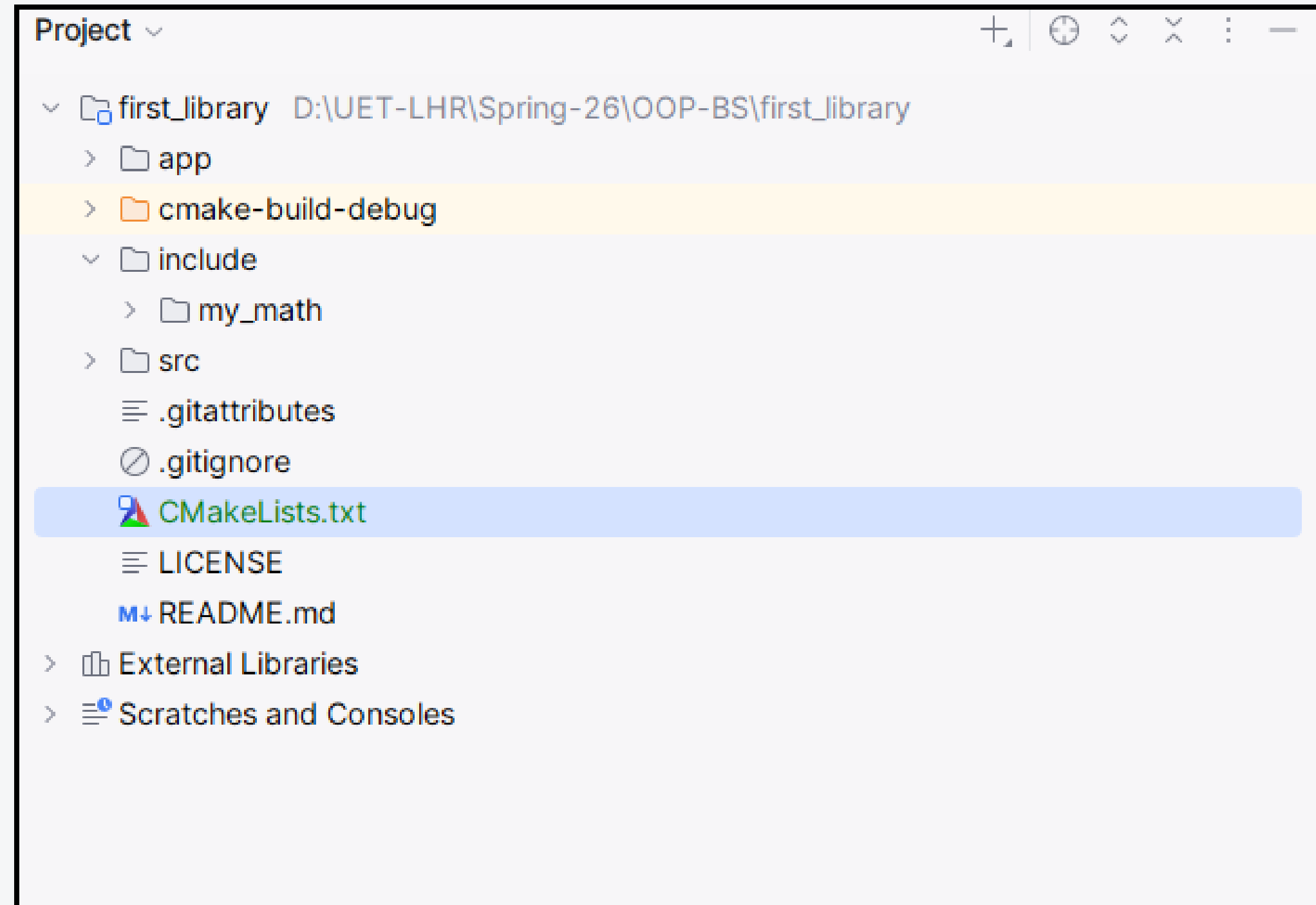
CREATING AND PUBLISHING A LIBRARY



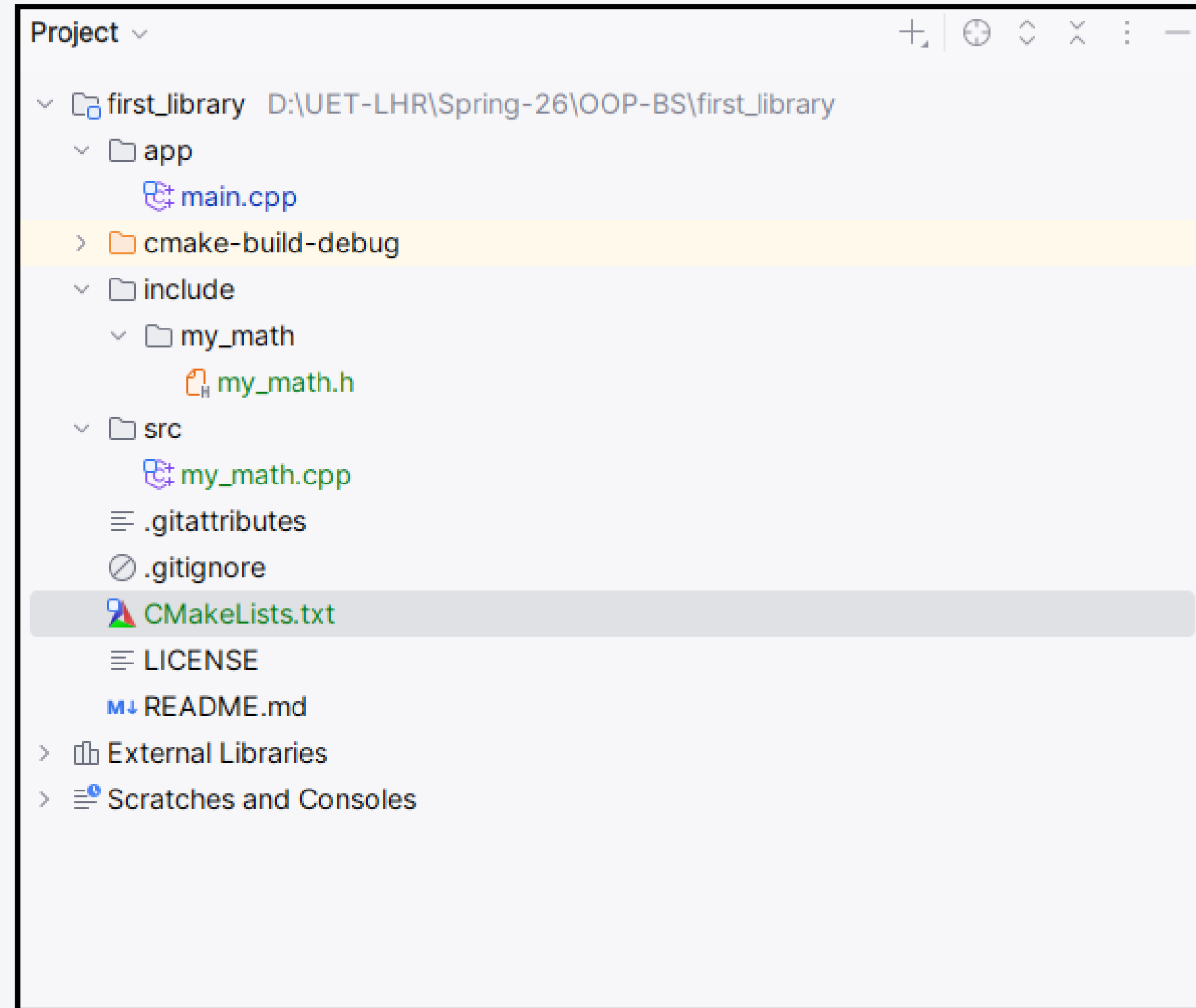
Dir Structure

This PC > New Volume (D:) > UET-LHR > Spring-26 > OOP-BS > first_library >				
Sort View ...				
Name	Date modified	Type	Size	
 .gitattributes	03/03/2026 10:02 pm	Git Attributes Sour...	1 KB	
 .gitignore	03/03/2026 10:02 pm	Git Ignore Source ...	1 KB	
 LICENSE	03/03/2026 10:02 pm	File	2 KB	
 README.md	03/03/2026 10:02 pm	MD File	1 KB	
 src	03/03/2026 10:20 pm	File folder		
 include	03/03/2026 10:20 pm	File folder		
 app	03/03/2026 10:20 pm	File folder		

Dir Structure



Dir Structure



Changes 8

History

Filter

8 changed files

✓.idea\gitignore

+

✓.idea\editor.xml

+

✓.idea\first_library.iml

+

✓.idea\modules.xml

+

✓.idea\vcs.xml

+

✓app\main.cpp

+

✓include\my_math_...\my_math_lib.h

+

✓src\my_math_lib.cpp

+

1

+

Default ignored files

2

+

/shelf/

3

+

/workspace.xml

4

+

Ignored default folder with query files

5

+

/queries/

6

+

Datasource local storage ignored files

7

+

/dataSources/

8

+

/dataSources.local.xml

9

+

Editor-based HTTP Client requests

10

+

/httpRequests/

Summary (required)

Description

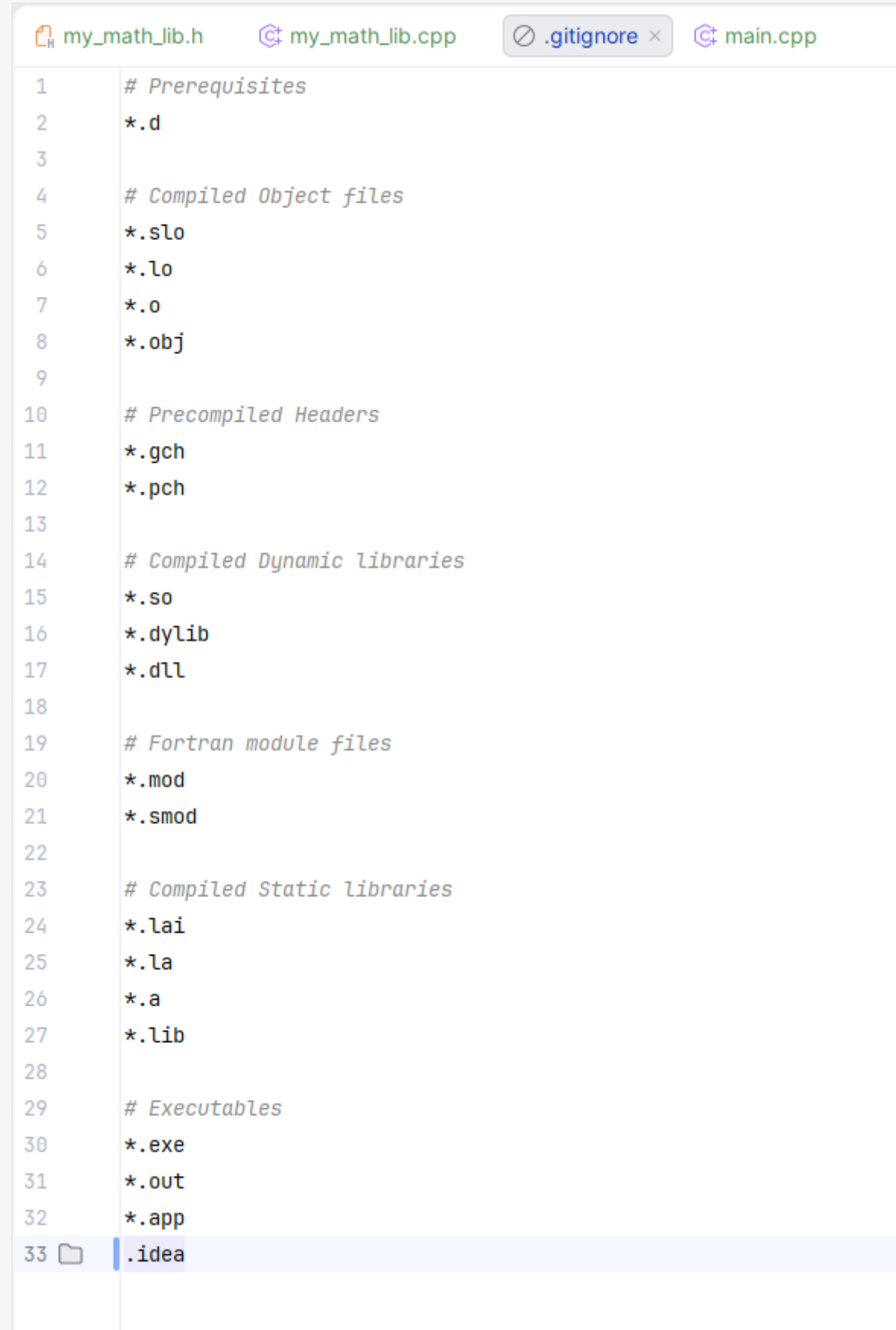
New

Commit 8 files to main

Committed 1 hour ago

Initial commit

Undo

gitignore

```
1  # Prerequisites
2  *.d
3
4  # Compiled Object files
5  *.slo
6  *.lo
7  *.o
8  *.obj
9
10 # Precompiled Headers
11 *.gch
12 *.pch
13
14 # Compiled Dynamic libraries
15 *.so
16 *.dylib
17 *.dll
18
19 # Fortran module files
20 *.mod
21 *.smod
22
23 # Compiled Static libraries
24 *.lai
25 *.la
26 *.a
27 *.lib
28
29 # Executables
30 *.exe
31 *.out
32 *.app
33  .idea
```

Commit

Changes 4

History

Filter

4 changed files

.gitignore

app\main.cpp

include\my_math_...\my_math_lib.h

src\my_math_lib.cpp

Summary (required)

Description

New

Commit 4 files to main

Committed 1 hour ago

Initial commit

Undo

↑

...

30

30

*.exe

31

31

*.out

32

32

*.app

✓ 33

+

.idea

Record the changes

Current repository
first_library

Current branch
main

↑

Push origin

Last fetched 5 minu...

1 ↑

Changes

History

Filter

0 changed files

No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.

Push commits to the origin remote

You have 1 local commit waiting to be pushed to GitHub.

Always available in the toolbar when there are local commits waiting to be pushed or `Ctrl + P`

Push origin

Open the repository in your external editor

Select your editor in [Options](#)

Repository menu or `Ctrl + Shift + A`

Open in Visual Studio Code

View the files of your repository in Explorer

Repository menu or `Ctrl + Shift + F`

Show in Explorer

Open the repository page on GitHub in your browser

Repository menu or `Ctrl + Shift + G`

View on GitHub

Summary (required)

Description

⧻

New

Commit to main

Committed just now
directories added.

Undo

Current repository
first_library

Current branch
main

Fetch origin
Last fetched just now

ChangesHistory

Filter

0 changed files

No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.

Open the repository in your external editor

Select your editor in [Options](#)

Repository menu or Ctrl + Shift + A

Open in Visual Studio Code

View the files of your repository in Explorer

Repository menu or Ctrl + Shift + F

Show in Explorer

Open the repository page on GitHub in your browser

Repository menu or Ctrl + Shift + G

View on GitHub

Summary (required)

Description

New

Commit to main

👤 shabansatti directories added.	a19eb32 · 1 minute ago	🕒 2 Commits
📁 app	directories added.	1 minute ago
📁 include/my_math_lib	directories added.	1 minute ago
📁 src	directories added.	1 minute ago
📄 .gitattributes	Initial commit	1 hour ago
📄 .gitignore	directories added.	1 minute ago
📄 LICENSE	Initial commit	1 hour ago
📄 README.md	Initial commit	1 hour ago

📖 README

📄 MIT license

first_library

This is first library repo.

About

This is first library repo.

- 📖 Readme
- 📄 MIT license
- 📈 Activity
- ★ 0 stars
- 👁 0 watching
- 🍴 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

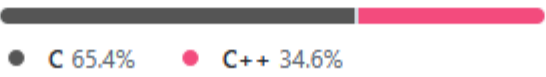
[Publish your first package](#)

Contributors 1

👤 shabansatti

Shaban Satti

Languages



Library Cpp

```
my_math.h  my_math.cpp x  .gitignore  main.cpp  CMakeLists.txt

1  #include "my_math/my_math.h"
2  namespace math {
3  →  int add(int a, int b) {
4      return a + b;
5  }
6  →  int subtract(int a, int b) {
7      return a - b;
8  }
9  →  int multiply(int a, int b) {
10     return a * b;
11 }
12 →  float divide(int a, int b) {
13     return a / b;
14 }
15 →  int modulo(int a, int b) {
16     return a % b;
17 }
18 } namespace math
19 |
```


Header File

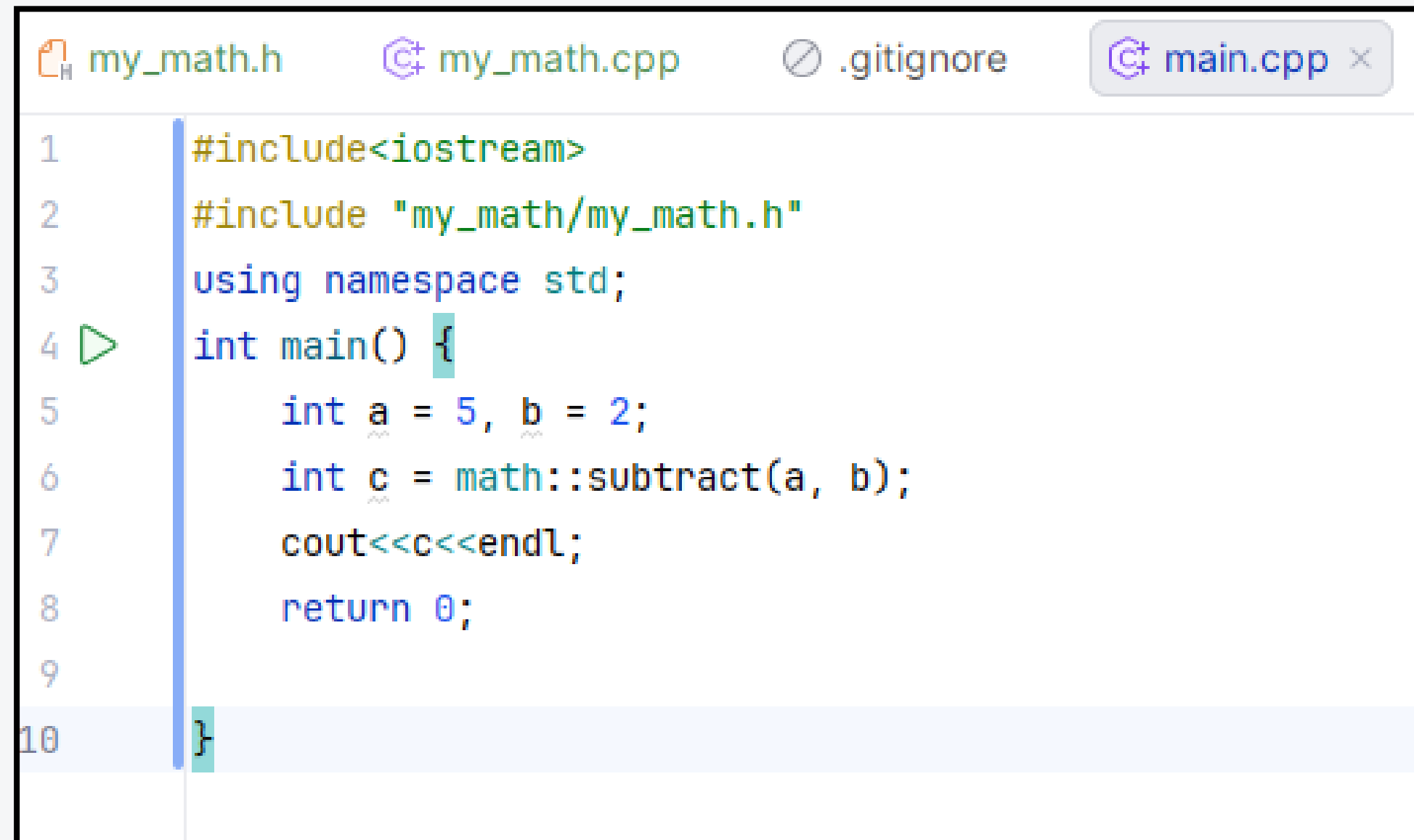


The screenshot shows a code editor with five tabs: my_math.h (active), my_math.cpp, .gitignore, main.cpp, and CMakeLists.txt. The active tab displays the following C++ header file content:

```
1  #ifndef FIRST_LIBRARY_MY_MATH_H
2  #define FIRST_LIBRARY_MY_MATH_H
3
4  namespace math{
5      int add(int a, int b);
6      int subtract(int a, int b);
7      int multiply(int a, int b);
8      float divide(int a, int b);
9      int modulo(int a, int b);
10 } namespace math
11
12 #endif
```

Line 7 has a tooltip that says "Apply Rename refactoring". Line 11 has a gear icon. The cursor is at the end of line 12.

App Code



```
my_math.h  my_math.cpp  .gitignore  main.cpp x
1  #include<iostream>
2  #include "my_math/my_math.h"
3  using namespace std;
4  int main() {
5      int a = 5, b = 2;
6      int c = math::subtract(a, b);
7      cout<<c<<endl;
8      return 0;
9
10 }
```

CMakeLists



```
1 cmake_minimum_required(VERSION 4.1)
2 project(first_library)
3 set(CMAKE_CXX_STANDARD 14)
4
5 add_library(my_math src/my_math.cpp)
6 target_include_directories(my_math PUBLIC ${PROJECT_SOURCE_DIR}/include)
7 add_executable(first_library app/main.cpp)
8
9 target_link_libraries(first_library PRIVATE my_math)
```

Current repository
first_library

Current branch
main

Fetch origin
Last fetched 25 minutes ago

Changes 6

History

app\main.cpp

Filter

6 changed files

app\main.cpp

CMakeLists.txt

include\my_math\my_math.h

include\my_math...\my_math_lib.h

src\my_math.cpp

src\my_math_lib.cpp

Summary (required)

Description

New

Commit 6 files to main

@@ -1,3 +1,10 @@

✓✓1

- //

✓✓2

- // Created by shaba on 03/03/2026.

✓✓3

- //

✓✓1

+ #include<iostream>

✓✓2

+ #include "my_math/my_math.h"

✓✓3

+ using namespace std;

✓✓4

+ int main() {

✓✓5

+ int a = 5, b = 2;

✓✓6

+ int c = math::subtract(a, b);

✓✓7

+ cout<<c<<endl;

✓✓8

+ return 0;

✓✓9

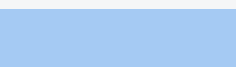
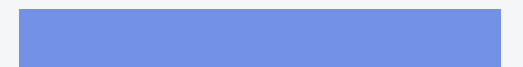
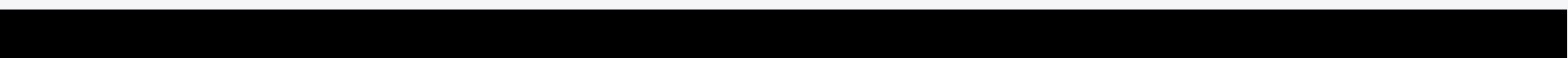
+

✓✓10

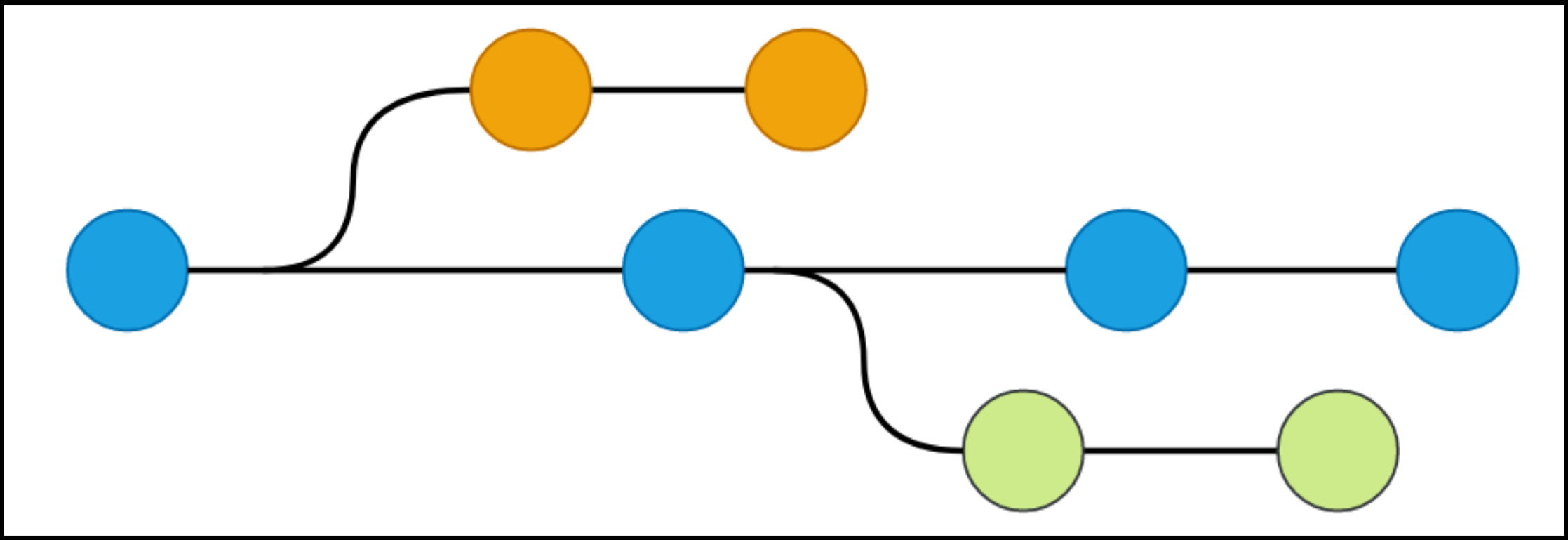
+ }

03.

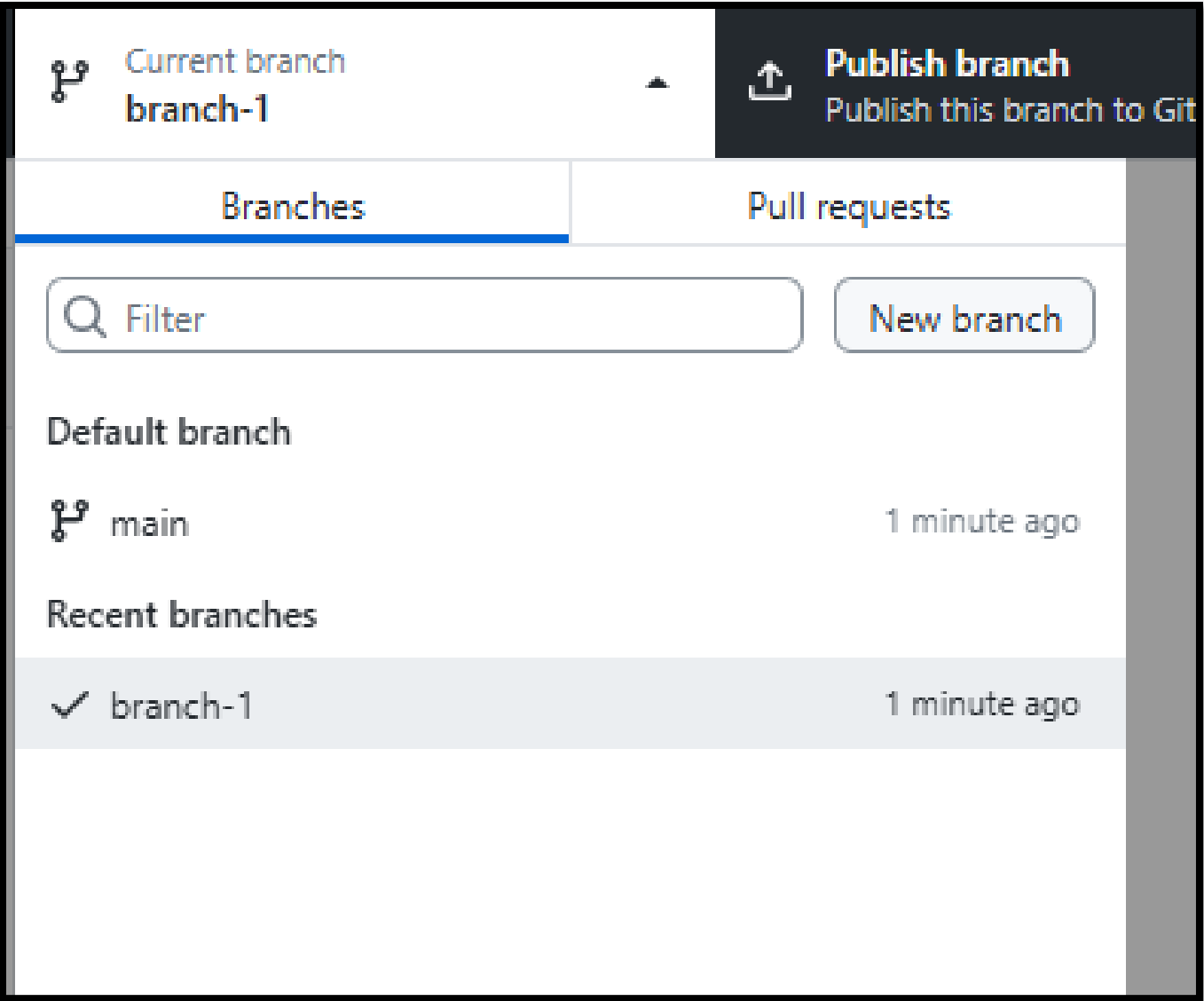
BRANCHING



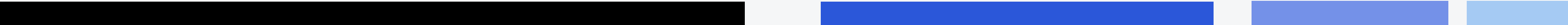
Branch

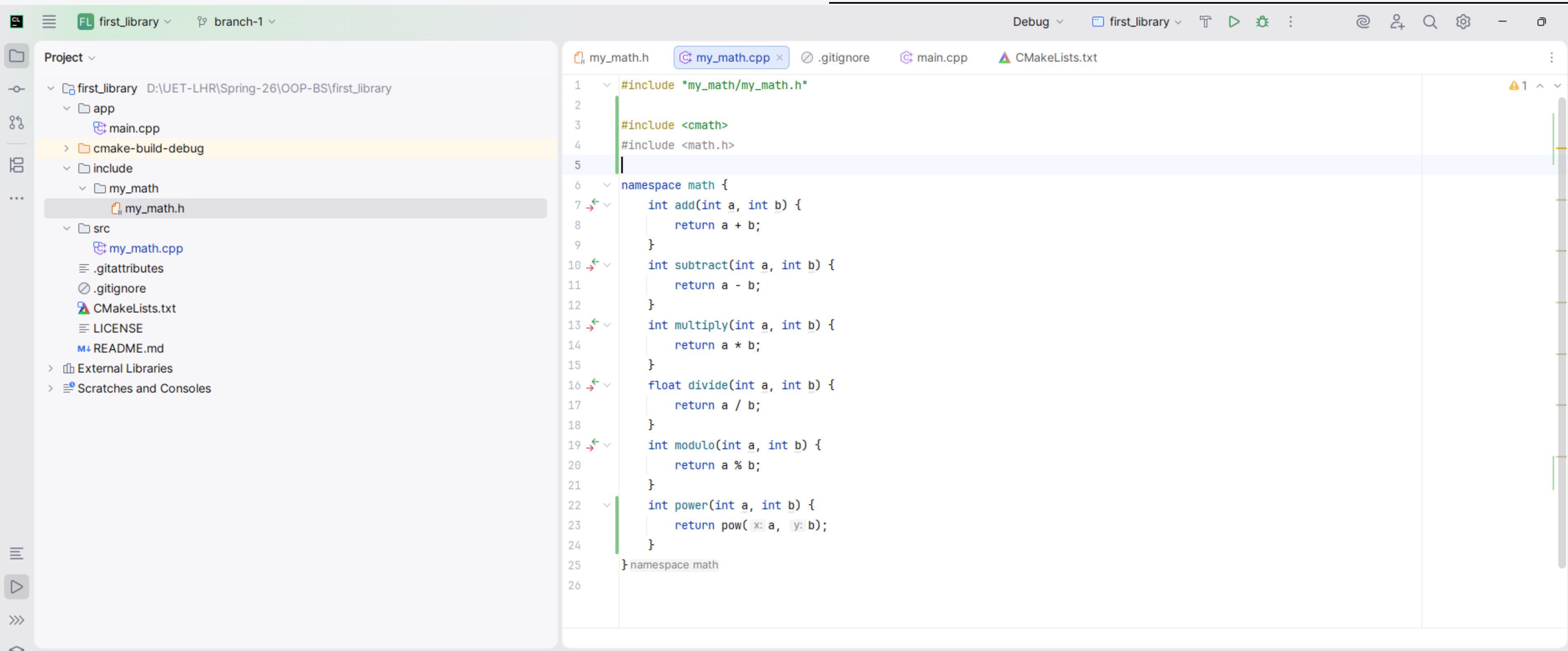


Branch



Now publish the branch. Check the online repo.





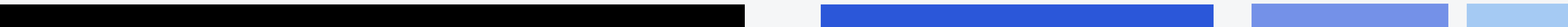
Branch



```
1  #ifndef FIRST_LIBRARY_MY_MATH_H
2  #define FIRST_LIBRARY_MY_MATH_H
3
4  namespace math{
5      int add(int a, int b);
6      int subtract(int a, int b);
7      int multiply(int a, int b);
8      float divide(int a, int b);
9      int modulo(int a, int b);
10     int power(int a, int b);
11 } namespace math
12
13 #endif #ifndef FIRST_LIBRARY_MY_MATH_H
```

Branch

Now commit the branch and push to origin.





first_library

Public

Pin

Watch

0

Fork

0

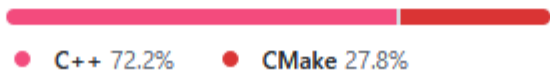
Star

0

main 2 Branches 0 Tags Go to file t Add file <> Code

A screenshot of a GitHub repository page. The repository name is 'first_library'. The description says 'This is first library repo.' The page shows the 'README' tab selected, with a link to the 'MIT license'. The repository has 0 stars and 0 forks.

● C++ 72.2% ● CMake 27.8%



Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

🔗

base: main ▾

⬅

compare: branch-1 ▾

✓ Able to merge. These branches can be automatically merged.

Add a title *

Branch 1

Add a description

Write

Preview

H B I ≡ <> 🔗 | ≡ ≡ ≡ | 📎 @ ↻ ↩ □

Add your description here...

📄 Markdown is supported

🖼 Paste, drop, or click to add files

Create pull request ▾

📘 Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

👤 2 commits

📄 3 files changed

👤 1 contributor

Reviewers

No reviews

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

[GitHub Community Guidelines](#)

shabansatti / first_library

🔍 Type / to search

<> Code

🕒 Issues

🔗 Pull requests

1

🔄 Actions

📁 Projects

📖 Wiki

🛡 Security

📊 Insights

⚙ Settings

New power function added. #1

Edit

<> Code

🔗 Open

shabansatti wants to merge 2 commits into `main` from `branch-1`

💬 Conversation 0

🔗 Commits 2

📄 Checks 0

📄 Files changed 3

+9 -1

shabansatti commented now

Owner

...

Additional functionality has been added.

😊

🔗 shabansatti added 2 commits 7 minutes ago

🔗 power added

3f14295

🔗 Update main.cpp

4a67055

🔗

✓ No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)

Add a comment

Write

Preview

H

B

I

≡

<>

🔗

≡

≡

🔊

📎

@

🗨

↩

📄

Add your comment here...

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

🔔 Subscribe

You're not receiving notifications from this thread.

PR

shabansatti commented now

Owner ...

Additional functionality has been added.

shabansatti added 2 commits 8 minutes ago

power added

3f14295

Update main.cpp

4a67055

Commit message

Merge pull request #1 from shabansatti/branch-1

Extended description

New power function added.

Commit email

shabansatti02@gmail.com ↕

Confirm merge Cancel

Still in progress? [Convert to draft](#)

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications Customize

Subscribe

You're not receiving notifications from this thread.

1 participant

Lock conversation

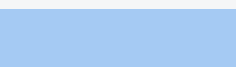
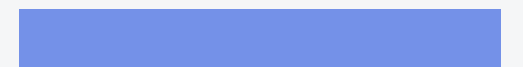
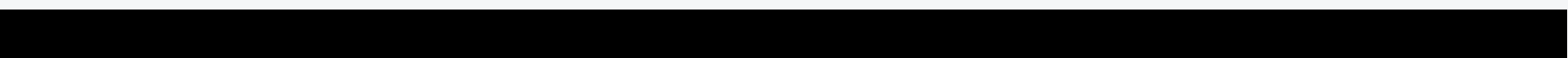
Add a comment

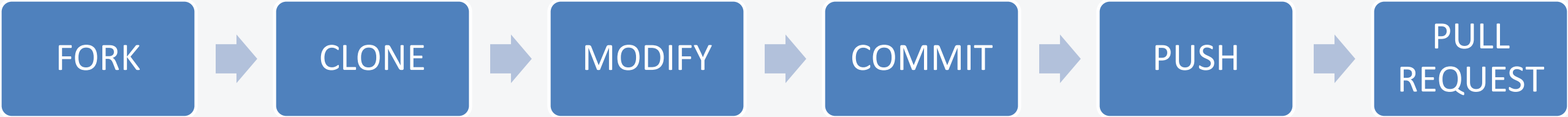
Write Preview

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04.

Working with other's Repos





S26-cpp-lab-01-364

Public

👁 Watch

🍴 Fork

0

★ Star

0

🔗 main ▾

🔗 1 Branch

🏷 0 Tags

🔍 Go to file

t

Add file

▾

<> Code

▾

Mian-Anas

Initial commit

7727140 · 2 weeks ago

🕒 1 Commit

📄 README.md

Initial commit

2 weeks ago

📖 README

✎

S26-cpp-lab-01-364

My name is Anas Ilyas.

About

My name is Anas Ilyas.

📖 Readme

📈 Activity

★ 0 stars

👁 0 watching

🔗 0 forks

Report repository

Releases

No releases published

Packages

Contributors

Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project.

Required fields are marked with an asterisk (*).

Owner *

shabansatti

▼

/

Repository name *

S26-cpp-lab-01-364

✔ S26-cpp-lab-01-364 is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

My name is Anas Ilyas.

22 / 350 characters

☒

Copy the

main

 branch only

Contribute back to Mian-Anas/S26-cpp-lab-01-364 by adding your own branch. [Learn more.](#)

ⓘ

You are creating a fork in your personal account.

Create fork

Fork

Repositories

My contributions

My repositories

My forks

Admin access

Views

Type / to search

+

New repository

owner:@me fork:true

2 repositories

Relevance

shabansatti/forked-repo

Public

Forked From Anas

1 • 0 • 0 • 0 • Updated 2 weeks ago •

shabansatti/Python

Public

My Python Examples

Python • MIT License • 13k • 0 • 0 • 0 • Updated on Aug 24, 2023 •



shabansatti / forked-repo

Code

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

QType / to search

+

forked-repo

Public

forked from [Mian-Anas/S26-cpp-lab-01-364](#)

main

1 Branch

0 Tags

Go to file

Add file

Code

This branch is up to date with [Mian-Anas/S26-cpp-lab-01-364:main](#).

Mian-Anas

Initial commit

README.md

Initial commit

README

S26-cpp-lab-01-364

My name is Anas Ilyas.

Local

Codespaces

Clone

HTTPSSSHGitHub CLI

https://github.com/shabansatti/forked-repo.git

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

About

Forked From Anas

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

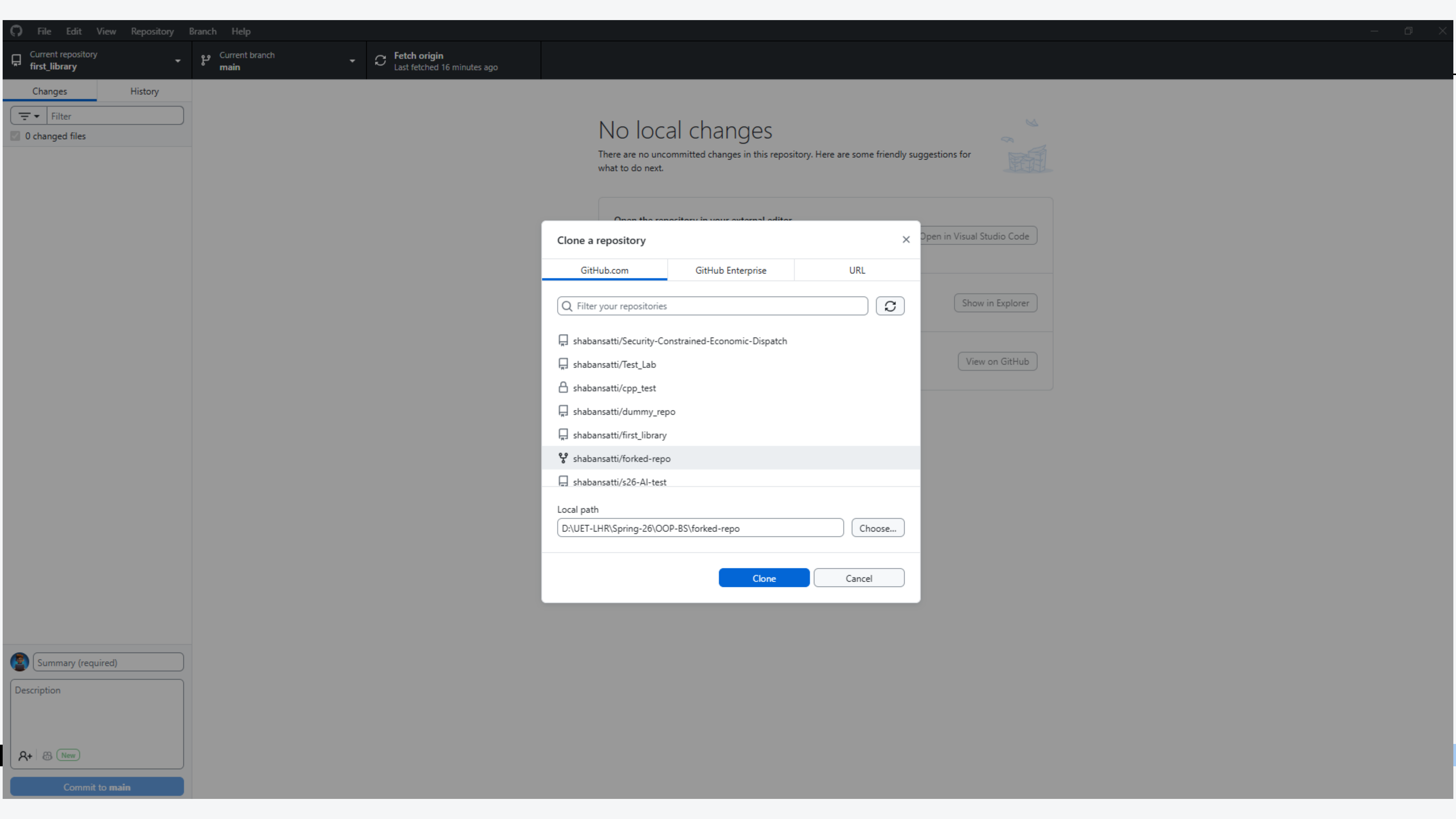
Packages

No packages published

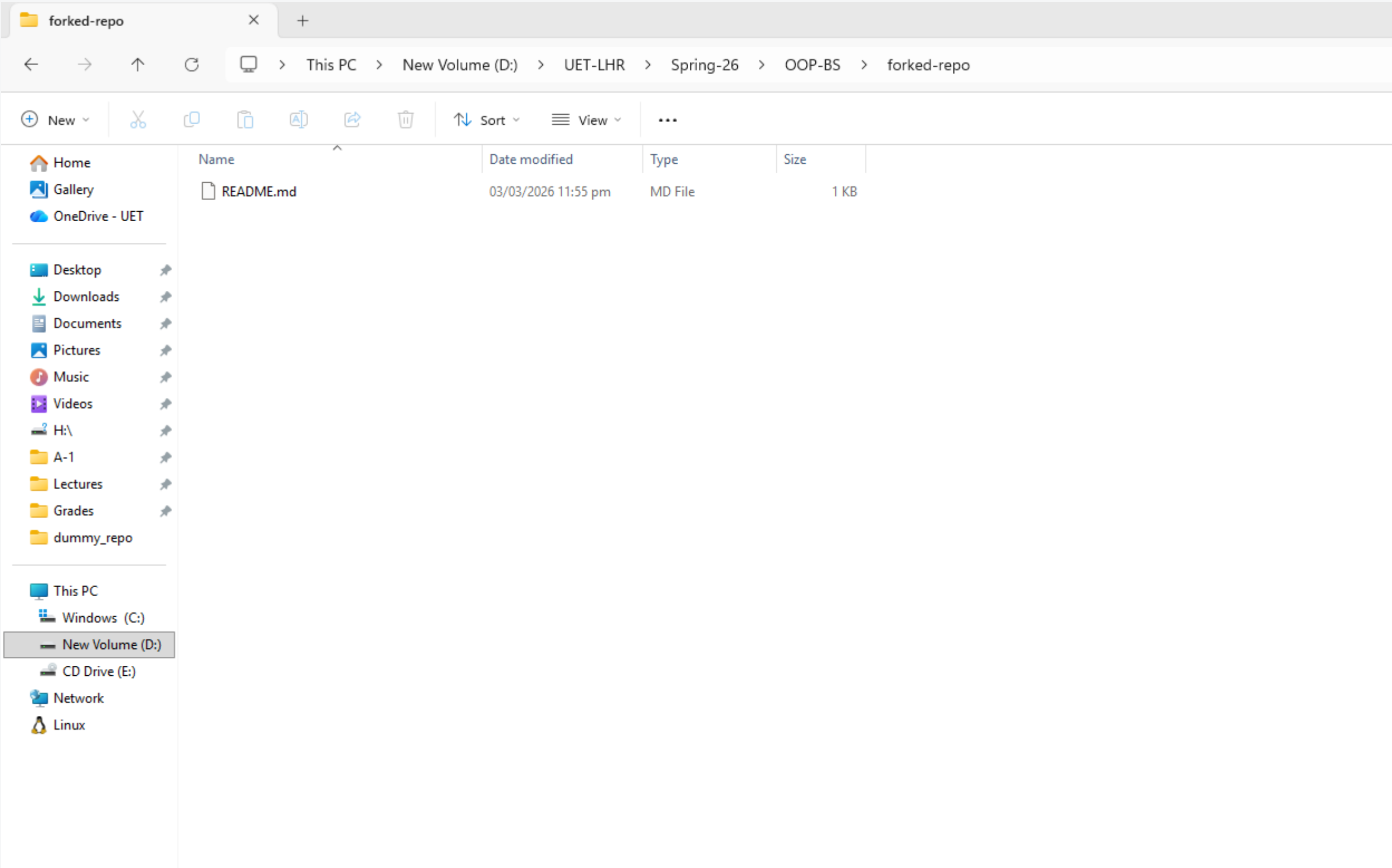
Publish your first package

Contributors

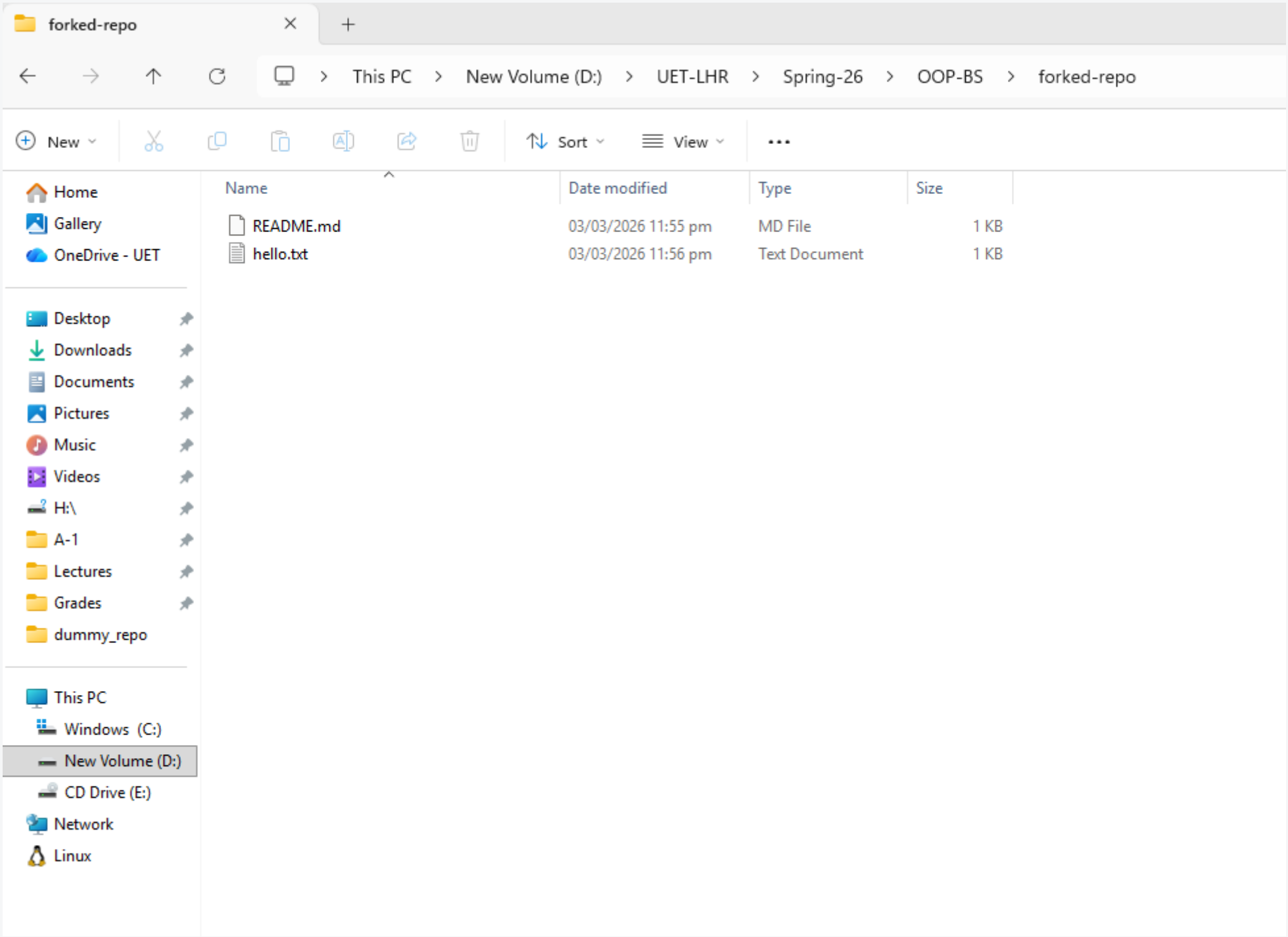
No contributors



Fork



Fork



shabansatti / forked-repo

Type / to search

<> Code

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

forked-repo

Public

forked from [Mian-Anas/S26-cpp-lab-01-364](#)

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

t

Add file

<> Code

This branch is 1 commit ahead of [Mian-Anas/S26-cpp-lab-01-364:main](#)

Contribute

Sync fork

shabansatti

created hello.txt

74d3d8c · now

2 Commits

README.md	Initial commit	2 weeks ago
hello.txt	created hello.txt	now

README

S26-cpp-lab-01-364

My name is Anas Ilyas.

About

Forked From Anas

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors

No contributors

forked-repo

Public

forked from [Mian-Anas/S26-cpp-lab-01-364](#)

📌 Pin

👁 Watch 0

🍴 Fork 0

★ Star 0

🔗 main

🔗 1 Branch

🏷 0 Tags

🔍 Go to file

Add file

<> Code

This branch is 1 commit ahead of [Mian-Anas/S26-cpp-lab-01-364:main](#).

🔗 Contribute

🔄 Sync fork

shabansatti

created hello.txt

📄 README.md

📄 hello.txt

This branch is 1 commit ahead of [Mian-Anas/S26-cpp-lab-01-364:main](#).

Open a pull request to contribute your changes upstream.

Open pull request

te ago

🕒 2 Commits

2 weeks ago

1 minute ago

📖 README

S26-cpp-lab-01-364

My name is Anas Ilyas.

About

Forked From Anas

📖 Readme

📈 Activity

★ 0 stars

👁 0 watching

🍴 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors

No contributors

**Thank
You!**

